Combinatorics

Independent diagnostic, two proof questions

Time allowed: 45 minutes **Total: 14 marks**

Candidate identifier: _____ **Date:** _____

Instructions. Attempt both questions independently. Show complete arguments, including all relevant cases and any final verification. No internet, calculator, AI tools or hints during the timed sitting. If unfinished, write down any useful progress. Stop the timed work when time is called.

Question 1 **[7 marks]**

In a competition with $n \geq 1$ players, each pair of distinct players has played exactly one match, with no draws. Prove that the players can be listed in an order $P_1, P_2, \ldots, P_n$ such that P_i defeated P_{i+1} for each $1 \leq i < n$.

Question 2 **[7 marks]**

A $3 \times n$ rectangular grid of unit squares is coloured using red and blue. A *monochromatic rectangle* consists of four squares at the intersections of two distinct rows and two distinct columns, all of the same colour. Find the smallest positive integer n such that *every* red-blue colouring of a $3 \times n$ grid contains a monochromatic rectangle. Prove both that your number always works and that one fewer column may fail.

After the sitting (not part of the timed assessment): Record approximate time spent on each question, whether either problem was familiar, and whether you worked beyond the limit. If you need more space, use the separate answer sheets.

Q1: _____ min Q2: _____ min Prior exposure: _____